

Precept 7 - Solutions

Exercise 1 (Eggs gone bad)

"Prince-Udon" has been making waves in the food critics circles lately and the famed celebrity chef Jordan Clamsey decides to pay a surprise visit. Jerry, the owner of the above-mentioned restaurant, is clearly flattered by such attention and is planning to impress the critic with a classic ramen with an egg. However, he shortly realizes in agony that each of the n eggs in the fridge will go bad uniformly at random at some point within an hour. Given that all eggs' expiry times are independent, what is the distribution of the time when Jerry is able to find at least one good egg? Find the expected value of this time and evaluate the failure rate.

Solution:

Let L be the length of time that Jerry can find a good egg out of n eggs, and L_i be the lifetime of i^{th} egg. One can think of L as the lifetime of a system with n parallel components, where each component's lifetime is given by L_i , $i = 1, \dots, n$. Note that for $t \in [0, 1]$, we have

$$F(t) = \mathbb{P}\{L \leq t\} = \mathbb{P}\{L_1 \leq t, \dots, L_n \leq t\} = \mathbb{P}\{L_1 \leq t\} \dots \mathbb{P}\{L_n \leq t\} = t^n,$$

by independence of the L_i and by the fact that each L_i is uniform on $[0, 1]$.

The expectation can be calculated by $\mathbb{E}L = \int_0^\infty \mathbb{P}\{L > t\} dt$ since $L \geq 0$. So, $\mathbb{E}L = \int_0^\infty \mathbb{P}\{L > t\} dt = \int_0^1 (1 - t^n) dt = \frac{n}{n+1}$.

Now, $H(t) = -\log(1 - F(t)) = -\log(1 - t^n)$, so the failure rate is

$$h(t) = \frac{dH(t)}{dt} = \frac{nt^{n-1}}{1 - t^n}.$$

We can see that in this problem, h is increasing:

$$\frac{dh(t)}{dt} = \frac{n(n-1)t^{n-2} + nt^{2n-2}}{(1 - t^n)^2} \geq 0.$$

Exercise 2 (Basics of Poisson processes and conditional distribution of arrival times)

Consider the following situation:

- A very dedicated ORF309 TA is answering last-minute questions via email from his/her dorm room the night before the exam, starting at 10 p.m., or $t = 0$.

- Equally dedicated ORF309 students are studying all night for the exam, starting at 10 pm ($t = 0$) until 10 am ($t = a$). They send emails to the TAs according to a Poisson process with rate λ .
- After each email, the TA decides to either stay up and wait for additional questions or go to bed.
- The TA considers the night a success if he/she goes to bed and no further questions arrive. The night is a failure if he/she goes to bed and another question does arrive OR the TA does not go to bed and no further questions arrive (in this case, the TA lost sleep for no reason).
- The TA decides to use the following policy: fix $s \in [0, a]$ beforehand and go to bed after the first email that arrives after s .

What is the probability of a successful teaching night for a fixed s ? What is the value of s that maximizes this probability and what is this optimal probability? Given that the night is a success, what is the distribution of the amount of sleep the ORF309 TA gets under policy s , assuming the alarm is set for $t = a$?

Solution:

Let N_t be the Poisson process representing email arrivals. Notice that

$$\begin{aligned}\{\text{successful night using policy } s\} &= \{\text{no emails after the first email after } s\} \\ &= \{\text{exactly one email after } s\} \\ &= \{N_a - N_s = 1\}\end{aligned}$$

By the properties of a Poisson process, $N_a - N_s \sim \text{Poisson}(\lambda(a - s))$ so

$$\begin{aligned}f(s) &:= \mathbb{P}\{\text{successful night using policy } s\} = \mathbb{P}\{N_a - N_s = 1\} \\ &= \lambda(a - s) e^{-\lambda(a - s)}.\end{aligned}$$

The derivative of this function is

$$\lambda e^{-\lambda(a - s)} (\lambda(a - s) - 1),$$

and we can conclude that the function $f(s)$ increases for $s \leq a - 1/\lambda$ and decreases for $s > a - 1/\lambda$ so it is optimal to choose $s^* = a - 1/\lambda$. The optimal probability of success is

$$f(s^*) = \lambda(1/\lambda) e^{-\lambda 1/\lambda} = 1/e.$$

Let X be the amount of time that the TA *stays awake* under policy s before a . Note that $s \leq X \leq a$. Let's compute

$$g(x) = \mathbb{P}(X \leq x \mid \{\text{successful night using policy } s\}).$$

Notice that we can restrict x to be in $[s, a]$ because $g(x) = 0$ for $x \leq s$ and $g(x) = 1$ for $x \geq a$.

$$\begin{aligned}\mathbb{P}(X \leq x \mid \{\text{successful night using policy } s\}) &= \mathbb{P}(X \leq x \mid N_a - N_s = 1) \\ &= \frac{\mathbb{P}\{X \leq x, N_a - N_s = 1\}}{\mathbb{P}\{N_a - N_s = 1\}} \\ &= \frac{\mathbb{P}\{N_x - N_s = 1, N_a - N_x = 0\}}{\mathbb{P}\{N_a - N_s = 1\}},\end{aligned}$$

because the email must have arrived before x . By the fact that Poisson processes have independent increments,

$$\begin{aligned} \frac{\mathbb{P}\{N_x - N_s = 1, N_a - N_x = 0\}}{\mathbb{P}\{N_a - N_s = 1\}} &= \frac{\mathbb{P}\{N_x - N_s = 1\}\mathbb{P}\{N_a - N_x = 0\}}{\mathbb{P}\{N_a - N_s = 1\}} \\ &= \frac{\lambda(x-s)e^{-\lambda(x-s)}e^{-\lambda(a-x)}}{\lambda(a-s)e^{-\lambda(a-s)}} \\ &= \frac{x-s}{a-s}. \end{aligned}$$

Therefore, $X \mid \{\text{successful night using policy } s\} \sim \text{Unif}(s, a)$. A simpler version of this is: given exactly one arrival before time t , the conditional distribution of the arrival is uniformly distributed between 0 and t (ask for intuition).

Finally, the question asked for the amount of sleep, so let $Y = a - X$ and we are in fact looking for $\mathbb{P}(Y \leq y \mid \{\text{successful night using policy } s\}) = \mathbb{P}(a - y \leq X \mid \{\text{successful night using policy } s\}) = 1 - g(a - y) = \frac{y}{a-s}$, so we have a conditional $\text{Unif}(0, a - s)$ distribution.

Exercise 3 (Poisson superposition and Bayes' theorem)

In a car repair shop wheels of the cars are inspected for safety and a repair/change is made in case it is necessary. The arrival process of cars requiring repairs of i wheels are independent Poisson processes with respective rates λ_i , where $\lambda_0 = 4, \lambda_1 = 2, \lambda_2 = 3, \lambda_3 = 2, \lambda_4 = 4$. The inspection time and the repair time of each wheel are both exponentially distributed random variables with mean 20 minutes. We can assume that the inspection time and the repair times of the wheels are independent of each other and of the customer arrival process. On a given day, it took 50 minutes to complete the job for the first car arrived to the shop. What is the probability that 2 wheels of that car needed a repair?

Solution:

Let (N_t^i) be the arrival processes of cars requiring i repairs. By the superposition theorem, N_t is a Poisson process with rate $\sum_{i=0}^4 \lambda_i = 15$.

Define X as the number of necessary wheel repairs for the first car. The thinning principle gives

$$\mathbb{P}\{X = 0\} = \frac{\lambda_0}{\lambda_0 + \dots + \lambda_4} = \frac{4}{15}$$

and similarly,

$$\mathbb{P}\{X = 1\} = \mathbb{P}\{X = 3\} = \frac{2}{15}, \quad \mathbb{P}\{X = 2\} = \frac{3}{15}, \quad \mathbb{P}\{X = 4\} = \frac{4}{15}.$$

Let T be the time spent for the first car. Then, given $X = 0$, the conditional distribution of T is $T \mid \{X = 0\} \sim \text{Expon}(\mu)$, where $\mu = \frac{1}{20}$. Moreover, given $X = i$, $T \mid \{X = i\} \sim \text{Gamma}(i + 1, \mu)$ as it is the sum of $i + 1$ independent identical exponential random variables. Using Bayes Theorem,

we can write

$$\begin{aligned}
\mathbb{P}\{X = 2|T = t\} &= \frac{\mathbb{P}\{T \in dt|X = 2\}\mathbb{P}\{X = 2\}}{\sum_{i=0}^4 \mathbb{P}\{T \in dt|X = i\}\mathbb{P}\{X = i\}} \\
&= \frac{\frac{\mu e^{-\mu t}(\mu t)^2 dt}{2} \mathbb{P}\{X = 2\}}{\sum_{i=0}^4 \frac{\mu e^{-\mu t}(\mu t)^i dt}{i!} \mathbb{P}\{X = i\}} \\
&= \frac{\frac{(\mu t)^2}{2} \mathbb{P}\{X = 2\}}{\sum_{i=0}^4 \frac{(\mu t)^i}{i!} \mathbb{P}\{X = i\}}
\end{aligned}$$

Upon plugging in the above probabilities and $\mu t = \frac{50}{20}$, we see that

$$\mathbb{P}\{X = 2|T = 50\} \approx 0.3115.$$